

I downloaded the data from the provided website, it has 699 cases.

I also understood this is a data science student kind of task, I have never done this previously, but I have all the ideas how to approach this.

So in this I found out there was 16 missing data's in the column 7 that is the bare nuclei.

So from this what is realized is I cannot use this data since it will reduce the accuracy, like when I am doing the mean calculation.

so I iterated through all the rows and where is found '?', I just removed that row entirely.

Next was to find the '1' in the column 2 clump thickness, and if the value was '1' i have removed the entire row, since I am not an expert in breast anatomy, I am not sure why, but there must be some reason.

Next was to make an weighted formula with 2 or more attributes except for column 1 and column 11 as we are gonna compare the column 11 and column 1 is the id of the data.

So for this I was researching how to calculate this I understood we need to average the attributes, I was also not sure which attribute to use (not an expert in the anatomy), so I came to a conclusion that I will average all the parameters except the 1 and 11, so the formula is simple (sum of all the parameters / total number of parameter). I have written some notes regarding this will attach below.

So after saving this in the CSV I was comparing the class and the mean, I found an interesting relation between both, when the mean value was less than 3 it was **benign** and when the value was greater than 3 it was **malignant**, which also kind a resembled with the value of the **2 = benign, 4 = malignant**.

So the next step was to find the correlation, so I was watching a youtube video where I got the idea of correlation, so I was looking at various methods of correlation and came down to 4 and I choose two of those methods because,

Spearman correlation: because we wanted to see whether higher predicted scores generally go with the higher class, without assuming a perfectly linear relationship

Point-biserial correlation: because the predicted score is continuous, while Class has only two groups: benign and malignant. Like binary 0/1.

so, I just used these two methods.

So how did I approach solving this:

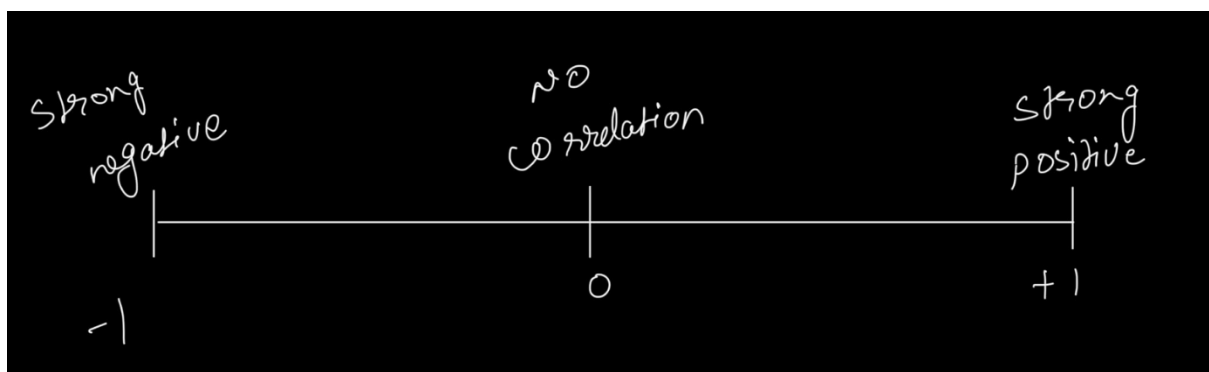
So first I was reading the data set and trying to understand what the file are it was bit overwhelming, also the extensions of the files were really confusing so I just opened the .data file I was in the form of CSV, so I just simply create a clone and just changed the extension to .csv, basically it's the same file diff extension so the excel software can detet the file, and opened there was no heading, of which column was what, so in the .name file they have given the heading so I just used the same heading order to make sense what the data was. So, I planned to use python, since, this task is unnecessarily complicated in c or cpp.

So the next task was to remove the rows with value 1 in the column 2 which was clump thickness. Next, I also wrote a code to remove the cells with no data, so when I searching through the csv file, I found out that only in the column 7 bare nuclei had '?', where the data was not available, so I have removed that row as well without the data, so removing the '?' in col 7 and '1' in col 1, the data sample went from 699 to 541 after filtering out the data.

After this is used the mean formula to calculate each row and the value was attached int the last row, column 12. And this was iterated did for all the 541 rows.

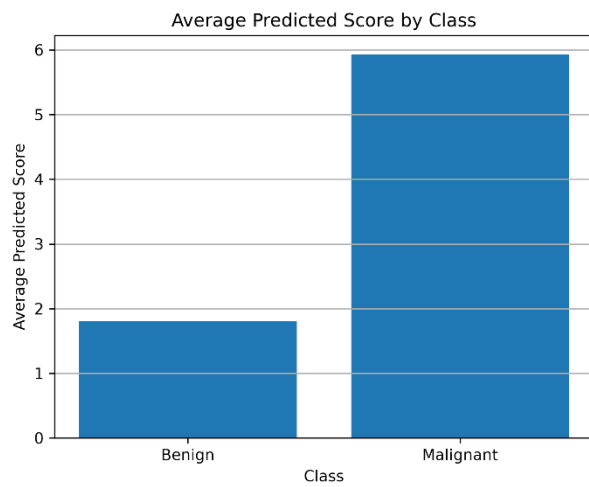
After this I was asked to do the correlation of the both the weight and the class, I just imported the lib, which is from scipy.stats import spearmanr, pointbiserialr, so this was the library and the module I was using since these two are the corelation method I wanted to use.

The spearman = 0.8482002870320052 and point-biserial = 0.8895973139680835, gave this as the results.



So both the values, are near the strong positive, this means the data says, there is a strong correlation between both the parameters.

Also, I added a graph, which is a bar graph, it showed malignant samples generally received higher predicted scores than benign samples. Actually, the graph is saved as a png.



I was mostly used the <https://docs.python.org/3.11/>, to look at the syntax and <https://docs.scipy.org/doc/scipy/tutorial/index.html#user-guide>, <https://docs.python.org/3/library/csv.html>, <https://matplotlib.org/stable/users/index.html> ., all these libs are open source.

So that was the python program.